The conjugate-pair family

The conjugate-pair family: 6 lines per generic F_{q⁴} curve, no descent, columns at chance level

2026-11-06. This is the third piece checking the claims that came from another session as an unverified assertion. The earlier pieces are `lift_dense_torsion_pairing_20261106` (char 2, 2-torsion pairing) and `lift_dense_orbit_parity_20261106` (odd char, criterion and descent).

The protocol was frozen before any data, at `b3954a0e9e` (`PROTOCOL.md`; post-hoc analysis under “Amendments”). The instrument is `conjpair/`, std-only Rust. Its field, curve and line code is copied from the previous piece’s checked `orbit/` crate. The run is `reports/run-20261106T0417Z/`, at commit `30a9b09fb5`. Report: `REPORT.pdf`.

Every statement below is an unverified assertion of this session [pcc:unverified]. The claim record is at the end.

The claim checked

> Two roots are Frobenius-conjugate. This needs n even, n ≥ 4, and full 2-torsion over K. There is no descent and the columns are useful. A generic such curve over F_{q⁴} has 6 of these lines, found by solving one F_q-linear equation. All 146 lines constructed behaved as predicted. Only 3 of about 1,050 points were dead, and the number of distinct columns was at chance level.

Construction

Take $E : y^2 = (x-e_1)(x-e_2)(x-e_3)$ over K = F_{q^n}, and a line $x_0 + u\cdot F_q$ with $t_i = (e_i - x_0)/u$. Pick a lone root c and a pair (a, b). Put w = 1/u and $x_0 = e_c - \gamma u$ with $\gamma \in K_0$ = F_{q^{n/2}}. Then the conditions “$t_c \in K_0$ and $t_b = \sigma^k(t_a)$” become one F_q-linear equation in (w, γ):

$(e_b - e_c)\cdot w + \gamma - \sigma^k(\gamma) = \sigma^k((e_a - e_c)\cdot w)$.

Its solutions modulo the shift (0, F_q) and F_q-scaling are the lines. With k ≠ n/2:

n	dimension of the solutions	lines
4	a point	3 × 2 = **6**
6	$P^1(F_q)$	12(q + 1)
2	none (only k = n/2 exists)	none

With k = n/2 a solution needs $N_{K/K_0}((e_b-e_c)/(e_a-e_c)) = 1$. That is not generic, and when it holds the root set is $\sigma^{n/2}$-stable, which is the descent case.

Answer

- **The family exists as claimed, on generic curves.**
  - **n = 4:** 31 of 32 curves with full 2-torsion over F_{13^4} and F_{101^4} have exactly 6 lines. Each comes from one F_q-linear system.
  - **n = 6:** all 4 curves over F_{13^6} have exactly 12(q + 1) = 168 lines, as derived.
  - **Behaviour:** every one of the 877 constructed lines (n = 4 and 6) satisfies the orbit criterion. Each has its lone root in K_0 and its pair in one Frobenius orbit, and each is all-lift or all-twist. **None** is a descent line ($\sigma^{n/2}$-stable), and no curve has $j \in K_0$. 0 exceptions.
  - **Share:** 449 of the 877 lines are all-lift (51 %). The others are all-twist.
- **The columns are useful at the scope measured** (n = 4, all-lift lines):

(p, n)	dead points	chance expectation	column-collision pairs	chance expectation
(13, 4)	10 / 689	7.6	75	81.1
(101, 4)	3 / 4,242	1.2	101	116.2

  - Descent lines in the previous piece had about half their points dead.
  - The planted positive control works on every line: adding P + T_c never adds a new column, so the detector does see 2-torsion pairing when it is there.
- **“6 lines” is the generic count, not a bound.**
  - One (13,4) curve has 19. On it, one system (and its mirror) has a 3-dimensional solution space instead of 2, which gives a pencil of q + 1 lines.
  - That curve has $j \notin K_0$, and all 19 of its lines pass every behaviour check.
  - This is a coincidence of probability about 1/q per system. It was 2 of the 384 n = 4 systems.
- **n = 2 has no conjugate-pair lines on generic curves.** One (101,2) curve with $j \in F_q$ (by chance) has k = n/2 solutions. That is the n = 2 descent case on a subfield-type curve. The raw count of 202 is solution points, which is about one line after scaling and shift (derived after the data).
- **Odd n.** The pattern needs the lone root in $F_{q^{n/2}}$, so odd n has no such lines. That is consistent with the previous piece, where no n = 3 line met the criterion except by chance.

What this does **not** establish:

- the cost: the 2^m yield per trial, against 6 lines per curve (n = 4) and their 51 % all-lift share;
- any scaling beyond q = 101.

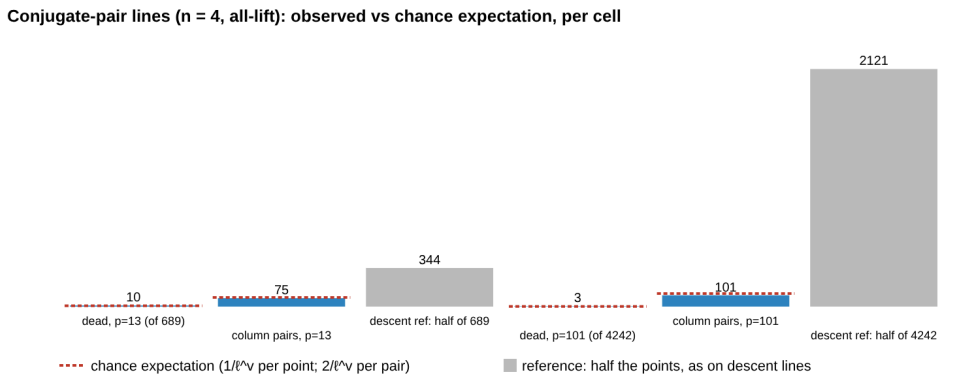
No ECDLP cost was measured, so there is no leaderboard, benchmark or rho row. This is the only family across the three pieces that survives as a candidate lever on generic curves.

Registered verdicts (`conjpair summarize`)

rule	verdict	detail
C1 n = 4: 6 lines on ≥ 90 % of curves, none > 6; k = n/2 gives 0 on ≥ 90 %	**FAIL**	31/32 with 6 lines, 1 curve with 19 (pencil); 32/32 with 0 from k = n/2
C2 behaviour of every constructed line	**PASS**	877 lines, 0 exceptions, 0 curves with j ∈ K_0
C3 n = 2: 0 lines	**FAIL**	1 of 16 curves (j ∈ F_q by chance) has k = n/2 solutions (202 raw solution points)
C4 dead ≤ E + 3√E + 3	**PASS**	10 vs 7.6 and 3 vs 1.2
C5, C5+ collisions ≤ E_2 + 3√E_2 + 3; planted control exact	**PASS**	75 vs 81.1 and 101 vs 116.2; control exact on 95/95 lines
C6 n = 6 count (report)	—	168, 168, 168, 168 against 12(q+1) = 168
Orders	ok	BSGS = enumeration on 16/16 (13,4) curves; [#E]P = O on all 52 curves
A/A	**PASS**	byte-identical `rows.jsonl` (sha256 `c083e110…`)

Per cell

p	n	lines per curve (k ≠ n/2)	null-space dimension per system
13	2	0 ×8	k = 1: dimension 1 on 48
101	2	0 ×8	k = 1: dimension 1 on 46, 2 on 2 (the j ∈ F_q curve)
13	4	6 ×15, 19 ×1	dimension 2 on 190, 3 on 2
101	4	6 ×16	dimension 2 on 288
13	6	168 ×4	dimension 3 on 120



dead points and collisions against chance

The full tables are in `SUMMARY.md`.

Sources checked

The `research-kb` server and CLI are not available in this container. A grep of the checkouts for this family found nothing. The external literature was not searched, as the source also says.

Reproduce

```
        cd conjpair && cargo build --release && cargo test --release
&& cd ..
        ./conjpair/target/release/conjpair run reports/run-X       #
~40 s, 4 threads
        ./conjpair/target/release/conjpair summarize reports/run-X
        ./conjpair/target/release/conjpair validate reports/run-X
claimed.json
```

Claim record

claim	standing	content
`PCC-20261106-707cc0`	not covered (unsigned)	line counts, behaviour, dead and collision tallies (`validator-report.json`, values in `claimed.json`)